Establishment-level life cycle and analysts' forecasts

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CES 26-12 **February 2026**

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Abstract

This paper examines how multi-unit firms' life-cycle stages affect analyst forecast accuracy. While prior studies focus on the firm-level life cycle, we utilize the Census data and focus on the establishment level. We find that analyst forecast accuracy is lower for multi-unit firms whose establishments are in different life-cycle stages than those in the same life-cycle stage. This finding suggests that the forecasting difficulty of more diversified firms can be attributed to the different life-cycle stages of each establishment. We also find that for firms whose units are in different stages, analyst forecast accuracy is lower if the establishments in earlier stages are larger (i.e., generate more revenue) than those in later stages. As a comparison, we estimate the life-cycle stages using firms' segment classifications in their 10-K filings. We find that analysts' forecast accuracy is lower when firms report fewer segments than the number of establishments, suggesting that aggregating more establishments for segment reporting could complicate analysts' forecasting. To our knowledge, this is the first study that focuses on the establishment-level life cycle. This study highlights that firm-level life cycles should not be taken without caution, as aggregating multiple units' life cycles may be misleading. In order to provide better forecasts to investors, analysts should have a deeper understanding of firms' subunits, especially when the establishments are in different life-cycle stages.

* Any views expressed are those of the authors and not those of the U.S. Census Bureau. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2479. (CBDRB-FY22-401) We are grateful for the guidance and assistance from Joseph Ballegeer at the Philadelphia RDC. We thank the financial support from Drexel University and the ESSEC Research Center. We thank the helpful comments from seminar participants at Drexel University. All errors are our own.

1. Introduction

Financial analysts gather information from various sources to make forecasts. For information gathering, analysts often visit plants to observe operations and facilities. By observing firm operations, analysts gain better insights into the firm's overall performance, enabling them to issue more accurate forecasts (Brown, Call, Clement, and Sharp 2015; Cheng, Du, Wang, and Wang 2016). However, forecasting becomes more difficult when firms are more diversified and have multiple operations (Duru and Reeb, 2002). In this study, we argue that the forecasting difficulty of more diversified firms can be driven by the different life-cycle stages of each unit. Specifically, this paper examines how multi-unit firms' establishment-level life-cycle stages influence analysts' forecast accuracy.[1]

Whereas most prior studies use the firm-level life cycle, this study focuses on the life cycle at the establishment level (i.e., the manufacturing plant level). Firm-level life cycle limits our understanding of the actual life cycle because the aggregation of multiple life cycles could drastically change the life-cycle stages. The firm-level life-cycle stage of a firm with two or more establishments could be difficult to analyze when all the establishments are in different stages of their own life cycles. By focusing on the establishment-level life cycle, investors can better understand the drivers of firm performance and more accurately predict future performance and profitability.

To our knowledge, this paper is the first to investigate the impact of different life cycles within the same firm. Specifically, we examine how multi-unit firms' life-cycle stages influence analysts' forecasts. Forecasting is likely to be easier for single-unit firms because analysts and

[1] We follow the Census Bureau's definition of a "multi-unit" firm. A multi-unit firm is defined as a firm that owns or operates two or more establishments but a single-unit firm is defined as a firm that has only one establishment (DeSalvo, Limehouse, and Klimek 2016). In the paper, we use units and establishments interchangeably.

investors need to forecast the performance of only one unit. However, it is more difficult for analysts to forecast multi-unit firms' performances. Moreover, if multi-unit firms' establishments are in different life-cycle stages, analysts could find it even harder to forecast overall performance. Prior studies show that the usefulness of accounting information varies across the life-cycle stages (Hribar and Yehuda, 2015; Dickinson et al., 2018). Thus, analysts should put different weights to various information depending on each establishment's life-cycle stage. Therefore, we predict that analysts' forecasts will be less accurate for multi-unit firms whose establishments are in different life-cycle stages than for multi-unit firms whose establishments are in the same life-cycle stage.

Next, we examine how the proportion of each life-cycle stage in multi-unit firms affects analysts' forecast accuracy. Prior studies show that the analysts' forecast accuracy varies across the life-cycle stages at the firm level. Hamers et al. (2016) show that analysts' forecasts tend to be less accurate during the non-mature stage due to low earnings persistence. Also, Hribar and Yehuda (2015) show that accruals and free cash flows are mispriced more in the growth stage. Because earlier stages in the life cycle are likely to have greater uncertainty in future earnings, we predict that analysts' forecasts will be less accurate if a greater proportion of a firm's establishments are in the earlier life-cycle stages.

To estimate the establishment-level life cycle, we use the confidential establishment data from the *Census of Manufacturers* (*CMF*) from the U.S. Census Bureau. For estimating the life cycle of establishments, we use six life-cycle descriptors at the establishment level and follow the method used by Anthony and Ramesh (1992) with some adjustments. That is, we rank establishments based on the descriptors and assign the establishments to five life-cycle stages. Our sample firms are the firms in the manufacturing sector between 1987 and 2017 that reported more than one establishment in their Census Bureau survey. We find that analysts' forecasts are less

accurate for multi-unit firms whose establishments are in different life-cycle stages than those in the same life-cycle stage, suggesting that the forecasting difficulty of more diversified firms can be attributed to the different life-cycle stages of each establishment. We also find that for firms whose units are in different stages, analyst forecast accuracy is lower if the establishments in earlier stages (i.e., the introduction stage and the growth stage) generate more revenue, have more capital expenditure, hire more employees, and have more production worker hours than the establishments in later stages.

For the additional analysis, we estimate the life cycles of segments using the 10-K segment data. We find that analysts' forecast accuracy is higher when firms have more disclosed segments. We also find that analysts' forecast accuracy is lower when firms report fewer segments than the number of establishments. These findings suggest that the 10-K segment disclosures provide useful information but aggregating more establishments for segment reporting could complicate analysts' forecasting.

This study makes several contributions. First, it contributes to the life cycle literature. To our best knowledge, this is the first study investigating different establishments' life cycles within the same firm. While previous life-cycle studies focus on firm-level life-cycle, we focus on more detailed Census establishment-level data, which enables us to break down one firm-level life cycle into multiple establishment-level life cycles within a firm. We aim to raise public awareness that a firm's life cycle stage may not be taken at face value. The perception of a firm's life cycle stage could result from aggregating multiple units and is thus misleading. Considering life-cycle stages at the establishment level can provide additional information about firms' operations, risks, and future performances.

Second, this study contributes to the analyst forecast literature. We identified another

determinant for analyst forecast accuracy and highlight the extra level of difficulty for the investors and information intermediates, such as financial analysts, to gather and process information of multi-unit firms. Better understanding subunits' different life cycles within a firm can improve analysts' forecast accuracy and facilitate market pricing.

The remainder of this paper is organized as follows. We review prior studies and develop hypotheses in Section 2. Research design is explained in Sections 3. We show empirical results in Section 4 and additional analyses in Section 5. We conclude in Section 6.

2. Hypotheses Development

Firm-level and Establishment-level Life Cycles

The original version of product life cycle has four stages: introduction, growth, maturity, and decline (Gort and Klepper 1982). In the introduction stage, a product is released into the market for the first time and, therefore, has high uncertainty and risks. Firms invest heavily in promoting the products and incur high costs. During this stage, the newly introduced product has low sales, but the product market is gradually growing. In the growth stage, product sales rapidly increase, and the market also grows fast. At the same time, competitors may emerge. In the maturity stage, firms face fierce competition, and the market becomes highly saturated. Thus, product sales and market growth are stagnant. In this stage, firms often develop or alter products to meet the demands of different market segments. Finally, in the decline stage, the product sales significantly drop, and the product's market share gradually decreases. The product then ultimately retires from the market.

Rather than examining product life cycles, prior studies concentrate on firm-level life cycles, which is an aggregation of multiple products' life cycles within the same firm. Firm-level

life cycle reflects a firm's evolution along distinct phases determined by internal factors, such as strategies or resources, and external factors, such as operating environment or macroeconomic conditions (Dickinson, 2011).

Some studies that focus on firm-level life cycles investigate firms' optimal strategies during different stages. Spence (1977) argues that firms can deter entry by incurring huge capital expenditures early in the life cycle. Spence (1979) rationalizes firms' early growth maximization because investments are irreversible and firms' ability to grow is limited. Other studies examine the usefulness of life cycle models in predicting returns and generating forecasts. Anthony and Ramesh (1992) show that the stock price reaction to accounting performance measures, such as sales growth and capital investment, is a function of the life-cycle stages. Vorst and Yohn (2018) find that a model based on a life-cycle stage leads to more accurate forecasts than other models based on economy-wide or industry-specific models. Dickinson, Kassa, and Schaberl (2018) show that investors assign more weight to analysts' forecasts during the growth and maturity stages but put more weight on accounting information during the introduction and decline stages. Hamers, Renders, and Vorst (2016) find that analyst forecasts tend to be more accurate for non-mature firms and that the accuracy increases when life cycles align between a firm and its industry. Focusing on mispricing in different stages, Hribar and Yehuda (2015) document the highest mispricing of both accruals and free cash flows in the growth stage.

Whereas prior studies focus on the firm-level life cycle, this study focuses on the establishment-level life cycle. One concern of firm-level life cycle measures used in the literature is that they might not correctly reflect the actual life cycle of a firm because the aggregation of units of different life-cycle stages may produce a misleading life-cycle stage measure for the firm. For example, a firm with two establishments may be incorrectly classified as a mature firm when

one establishment is in the growth stage and the other is in the decline stage.

Establishment-level life cycle measures can mitigate such problems for the following reasons. First, the establishment-level life cycle measures better capture the product life cycles than the firm-level life cycles do. While some establishments could produce and sell multiple products, more similar products are likely produced at each establishment. However, it is difficult to identify different types of products that lead to different life cycles using a firm-level life cycle because firm-level life cycles are determined after aggregating all the products the firm produces and sells.

Second, each establishment likely has a life cycle that differs across establishments within the same firm. One establishment's life-cycle stage is not always the same as the other establishment's life-cycle stage. Therefore, analyzing the life cycles at the establishment level enables us to better understand the actual product life cycles. It makes it easier for investors and analysts to identify similar operations and businesses that contribute to distinct life cycles within the same firm.

Establishment-level life cycles and analysts' forecast accuracy

Prior studies document that greater diversification leads to greater forecasting difficulties for analysts. Duru and Reeb (2002) find that international diversification is associated with less accurate and more optimistic analysts' forecasts. They argue that this forecasting difficulty stems from firms' more volatile foreign earnings, analysts' less-specialized knowledge of foreign markets, and managers' greater discretion in foreign operations. We argue that it is more difficult for analysts to forecast multi-unit firms because different units in the same firm may be at different life-cycle stages.

In the same spirit, analysts may find it more difficult to forecast the performances of multiple operations of diversified firms. Forecasting is relatively easier for single-unit firms because analysts need to forecast the performance of only one establishment. Moreover, if multi-unit firms' establishments are in different life-cycle stages, it could be even more difficult for analysts to forecast the firms' overall future performance. Prior studies show that the information embedded in earnings components and the relevance of earnings vary across the life-cycle stages (Hribar and Yehuda, 2015; Dickinson et al., 2018). Dickinson et al. (2018) find that investors value accounting information more during the introduction and decline stages. Thus, for multi-unit firms that have multiple life cycles, it would be difficult for analysts to extract and analyze the different signals and information depending on each establishment's life-cycle stage. However, analysts could find it relatively easier to forecast multi-unit firms when all the establishments of a firm are in the same life-cycle stage. In this case, analysts can focus on the most relevant information in a certain life-cycle stage, leading to more accurate forecasts. Therefore, we propose the first hypothesis as follows.

H1: Analysts' forecast accuracy is lower for multi-unit firms whose units are in different life-cycle stages than for multi-unit firms whose units are in the same life-cycle stage.

While the establishment-level life cycle could help analysts more accurately delineate a multi-unit firm's overall performance, the proportion of establishments in each life-cycle stage could also affect analysts' forecast accuracy. Prior studies document that analysts' forecast errors are associated with greater uncertainty in future earnings (Barth and Hutton 2004; Dichev and Tang, 2009; Donelson and Resuteck, 2015). The uncertainty in future earnings vary across the life-cycle stages. Hamers et al. (2016) show that analysts' forecast accuracy is likely to be low during the

non-mature stage due to low earnings persistence. Also, Hribar and Yehuda (2015) show that the growth stage is associated with greater mispricing in accruals and free cash flows. Because earlier stages in the life cycle are likely to have greater uncertainty in future earnings, we predict that analysts' forecasts will be less accurate if more establishments of a firm are in the earlier life-cycle stages. Thus, we present our second hypothesis as follows.

H2: Analysts' forecast accuracy is lower if a greater proportion of a multi-unit firm's establishments are in the earlier life-cycle stages.

3. Research design

Sample

We use the confidential establishment data from the *Census of Manufacturers* (CMF) and the *Longitudinal Business Database* (LBD) from the U.S. Census Bureau.[2] The CMF is conducted every five years as a part of the Economic Census, which is the U.S. Government's official five-year measure of the U.S. business and economy. The surveys for the Economic Census are authorized by Title 13, United States Code (U.S.C.) (sections 131, 224, and 225), which requires establishments to answer the survey questions and return them to the U.S. Census Bureau. Firms that fail to report or deliberately provide false information are subject to penalties by the law.

To estimate the establishment-level life cycle, we collect the total value of shipments (*TVS*), capital expenditure (*CAPEX*), employment (*EMP*), salary and wages (*WAGE*), and production worker hours (*PRODHR*) from CMF at the establishment level every five years from 1982 to 2017.

[2] We use the *Census of Manufacturers* (CMF) rather than the *Annual Survey of Manufacturers* (ASM) because of the greater coverage of establishments in the CMF than the ASM. While only large establishments tend to participate in the ASM, most establishments receive and respond to the CMF except for very small establishments that have less than four percent of all sales and receipts nationally. See the U.S. Census Bureau website (http://www.census.gov) for more details.

We also extract establishment age (*AGE*) from the LBD every year. Our sample year starts from 1987 because we need two consecutive CMF years to calculate the growth rates of the CMF variables. For example, we compute the five-year growth rate in 1987 by using 1982 and 1987 data.

Since the CMF is conducted for manufacturing establishments, we only use firms in the manufacturing sector (i.e., SIC code between 2000 and 3999) in the U.S. Because the NAICS began to supplant the SIC in *CMF* in 2002, we use concordance tables from the Census Bureau that provides the matching between SIC and NAICS codes.

We also exclude observations that have poor matching between the Census Bureau's LBD and Compustat. Following Bens et al. (2011), we exclude observations when the ratio of the firm's total value of shipments (*TVS*) in LBD to the firm's sales in Compustat is less than 0.75 or greater than 1.25.[3]

In addition to the Census Bureau data, we use Compustat for financial statement data, CRSP for stock data, and I/B/E/S for analysts' forecast and management forecast data. From I/B/E/S, we use the individual analysts' annual EPS forecasts. If an analyst makes multiple forecasts for the same firm on the same day, we only keep the last forecast. We also exclude firms that have only one establishment because we need multiple establishments per firm to estimate the different life cycle stages for each establishment. Lastly, we exclude observations with missing Compustat, CRSP, or I/B/E/S variables. The final sample has 63,500 analysts' EPS forecasts and 2,000 firm-year observations between 1987 and 2017.[4]

[3] Bens et al. (2011) argue that when the ratio is less than 0.75, the firm is likely to have non-manufacturing sales or generate sales from plants outside the U.S, and that when the ratio is greater than 1.25, it is likely that the firm's intersegment sales are not removed by the Census Bureau.

[4] To protect confidential data, the Census Bureau requires rounding of all statistical output. Accordingly, the number of observations in the paper follows the rounding rules. The specific rounding rules for the actual number of observations are as follows: if N is between 100 and 999, round to the nearest 50; if N is between 1,000 and 9,999, round to the nearest 100; and if N is between 10,000 and 99,999, round to the nearest 500.

Estimation of establishment-level life cycles

Our estimation of the establishment-level life-cycle stage follows the method used by Anthony and Ramesh (1992) with some modification of the life-cycle descriptors. To estimate the life-cycle stage of establishments, we use six life-cycle descriptors: sales growth, capex growth, employment growth, growth in salary and wages, growth in production worker hours, and firm age at the establishment level.[5]

Prior studies argue that physical and financial factors are associated with the growth of firms and that early-stage and growing firms make early investments in these factors (Spence 1977, 1979, 1981; Jovanovic, 1982). Similar to the expectations in Anthony and Ramesh (1992), our expectations are as follows. Establishments in the early life-cycle stages are expected to show higher growth in sales. Early-stage establishments are likely to increase investments in equipment and facilities, leading to higher growth in capital expenditures. Growing establishments need more workers and make investments in high-quality employees, leading to higher growth in employment and higher growth in salary and wages. As establishments' sales increase, production workers are expected to work longer hours to keep up with rising demand, leading to higher growth in production worker hours. The growth rates of these five descriptors are expected to be smaller towards the decline stage. Lastly, the age of establishments will be younger for the early stages and will be older towards the decline stage.

We estimate the life-cycle stages based on six variables: (1) the growth rate of the total value of products shipped (*GTVS*), (2) the growth rate of the capital expenditure for depreciable

[5] Anthony and Ramesh (1992) estimate the firm-level life-cycle stages using four variables (dividend payout as a percentage of income, sales growth, capex as a percentage of the total firm value, and firm age) as indicators of the life-cycle stage. Since establishments do not have dividend payout, we exclude this variable. We add employment growth, growth in salary and wages, and growth in production worker hours.

assets (*GCAPEX*), (3) the growth rate of employment (*GEMP*), (4) the growth rate of salary and wages (*GWAGE*), (5) the growth rate of production worker hours (*GPRODHR*), and (6) the establishment age (*AGE*). We define the first five variables as follows.

$$GTVS_t = (TVS_t/TVS_{t-5})\times 100$$
$$GCAPEX_t = (CAPEX_t/CAPEX_{t-5})\times 100$$
$$GEMP_t = (EMP_t/EMP_{t-5})\times 100$$
$$GWAGE_t = (WAGE_t/WAGE_{t-5})\times 100$$
$$GPRODHR_t = (PRODHR_t/PRODHR_{t-5})\times 100$$

where

TVS_t = The establishment's total value of products shipped in year t.
$CAPEX_t$ = The establishment's capital expenditure for depreciable assets in year t.
EMP_t = The establishment's total employment in year t.
$WAGE_t$ = The establishment's total salary and wages in year t.
$PRODHR_t$ = The establishment's total production worker hours in year t.

While *AGE* is a yearly variable as we can calculate it from the *Longitudinal Business Database* (LBD), *TVS*, *CAPEX*, *EMP*, *WAGE,* and *PRODHR* are available in the *Census of Manufacturers* (CMF) every 5 years. For *GTVS*, we use the most recent *TVS* value for estimating the life-cycle stages every year. For example, we collect *TVS* in 2007 and 2012 from the CMF to calculate $GTVS_{2012}$ and use this value from 2008 to 2012 for estimating life-cycle stages. We do the same calculation for the other growth variables. For life cycle estimation, using the data of a 5-year interval will not significantly influence the results because the life-cycle stages do not change rapidly but change slowly over a long period.

Following Dickinson (2011), we use the six life-cycle variables using the Census data to group establishments into five life-cycle stages (introduction, growth, maturity, shake-out, and decline).[6]

[6] The five life-cycle stages are explained in Dickinson (2011) with economic theories. See Table 1 in Dickinson (2011). Although we use five stages in the main analysis, we get similar results with three life-cycle stages (growth, maturity, and decline).

Specifically, we use a method like that employed by Anthony and Ramesh (1992) which use rankings of the descriptors. We assign scores to establishments based on the rankings of the six variables. For the age variable, we divide the establishments into Young, Adult, and Old groups. For *GTVS*, *GCAPEX*, *GEMP*, *GWAGE*, and *GPRODHR*, we categorize the establishments into Low, Medium, and High groups. Once an establishment-year is assigned to a group, we assign a score to each group. The first group (Young or High) will receive a score of 1. The second group (Adult or Medium) will receive a score of 2, and the last group (Old and Low) will receive a score of 3. We create five life-cycle groups based on a composite score obtained by averaging the scores for six variables. To balance the number of observations in each stage group, establishments are equally assigned to five life-cycle groups based on the ranking of the composite score.[7] Establishments in five groups represent the following stages: the first group = introduction stage, the second group = growth stage, the third group = maturity stage, the fourth group = shake-out stage, and the fifth group = decline stage.

Analysts' forecast accuracy measure

We measure analysts' forecast accuracy ($ACCURACY_{i,j,t}$) as the negative of the absolute difference between analyst *j*'s forecast of firm *i*'s forecasted earnings and actual earnings, divided by the stock price at the end of the previous year for firm *i* in in year *t*:

$$ACCURACY_{i,j,t} = (-1)\left|FORECAST_{i,j.t}^{i,j,t-1} - ACTUAL_{i,t}\right|/PRICE_{i,t-1} \qquad (1)$$

where $FORECAST_{i,j,t}^{i,j,t-1}$ is analyst *j*'s forecast of firm *i*'s period *t* earnings made at period *t-1*, $ACTUAL_{i,t}$ is firm *i*'s actual earnings per share for period *t*, and $PRICE_{i,t-1}$ is the stock price at

[7] Alternatively, we estimate life-cycle stages using the composite-score rankings of establishments in the same 4-digit SIC code and get similar results (untabulated).

the end of year *t-1*. We take the negative of the forecast error so that a higher value of *ACCURACY* indicates a more accurate analysts' consensus forecast.

Testing H1

To examine whether analysts' forecast accuracy is lower for multi-unit firms whose units are in different life-cycle stages than for firms whose units are in the same life-cycle stage, we will use the following model:

$$\begin{aligned} ACCURACY_{ijt} = {} & \beta_0 + \beta_1 SAMELC_{it} + \beta_2 ESTABNUM_{it} + \beta_3 SAMELC * ESTABNUM_{it} \\ & + \beta_4 SIZE_{it} + \beta_5 EVOL_{it} + \beta_6 SKEW_{it} + \beta_7 LOSS_{it} + \beta_8 BTM_{it} + \beta_9 DISP \\ & + \beta_{10} FOLLOW_{it} + \beta_{11} \Delta EARN_{it} + \beta_{12} MFNUM_{it} + \beta_{13} A_EXP_{ijt} + \beta_{14} B_ANA_{ijt} \\ & + \beta_{15} NoSIC2_{ijt} + \beta_{16} NoFirms_{ijt} + \varepsilon_{ijt} \end{aligned} \quad (2)$$

The dependent variable is analysts' forecast accuracy ($ACCURACY_{ijt}$). We use an indicator variable, $SAMELC_{it}$, which is equal to one if all the establishments of a multi-unit firm *i* are in the same life-cycle stage in year *t* and zero otherwise. For example, if a firm has all its establishments in the growth stage, $SAMELC_{it}$ receives a value of one. However, if a firm comprised of three establishments has one establishment in the growth stage and two establishments in the maturity stage, $SAMELC_{it}$ is assigned a value of zero because at least one unit is in a different life-cycle stage estimation. *SAMELC* is constructed based on the life-cycle estimation using the composite-score rankings of all establishments as explained in Section 3.2. For H1, we expect to find a significantly positive β_1 if firms that have all their units in the same life-cycle stage are more likely to have more accurate analyst forecasts than are firms that have their units in different life-cycle stages.

We control for the degree of diversification by adding $ESTABNUM_{it}$ which is the natural logarithm of the number of the firm *i*'s establishments in year *t*. We expect to find less accurate analysts' forecasts when firms have more establishments. We also interact $SAMELC_{it}$ and

ESTABNUM$_{it}$ and expect to find a negative coefficient on this interaction term if analysts' forecasts are more accurate when multi-unit firms have few establishments and when all the establishments are in the same life-cycle stage.

We control for other factors that affect forecasting difficulty. We control for firm size (*SIZE*), which is the natural log of total assets. On the one hand, larger firms are associated with greater complexity, attributed to forecasting difficulty. On the other hand, larger firms are more likely to provide more information to the public or external information providers (e.g., media, analysts) generate more information for larger firms. Thus, the effect of firm size on analysts' forecast accuracy is unclear ex-ante.

Because analysts' forecast accuracy is associated with earnings skewness and changes in earnings (Lang and Lundholm, 1996; Gu and Wu, 2003), we will add *SKEW* and *ΔEARN*. Gu and Wu (2003) find that skewed earnings tend to have more optimistic forecasts. *SKEW*, which uses the past ten years of earnings, is defined as $\{n_i \sum_t((x_{it} - \bar{x}_i)/ s_i)^3\}/\{(n_i - 1)(n_i - 2)\}$ where x_{it} is the actual earnings per share for year *t* and firm *i*, $\bar{x}_i$ is the mean, s_i is the standard deviation of x_{it}, and n_i is the number of observations for firm *i*. *ΔEARN*, which measures the changes in earnings, is the absolute value of the difference between the current year's earnings per share and the last year's earnings per share, divided by the price at the end of the previous year. Prior studies show that firms with greater volatility in earnings tend to have less accurate and more optimistic analysts' forecasts (Lim, 2001). We also control for earnings volatility (*EVOL*), which is the standard deviation of return on assets over the previous five years.

To control for the uncertainty related to operating losses, we will include an indicator variable (*LOSS*) that is equal to one if firm *i*'s earnings are negative in year t and zero otherwise. Analysts tend to issue more optimistic forecasts when firms are experiencing losses (Brown, 2001).

We expect that analysts' forecast accuracy will be less accurate and that analysts will issue more optimistic forecasts. We also add the book-to-market ratio (*BTM*) to control for the growth opportunities that could affect the analysts' forecasting difficulty.

Prior studies show that the number of analysts following the firm and forecast dispersion are associated with forecast accuracy (Lys and Soo, 1995; Lang and Lundholm, 1996; Gu and Wu, 2003; Duru and Reeb, 2002; Hope, 2003). Thus, we will control the number of analysts following the firm (*FOLLOW*) because it could indicate the intensity of analysts' competition. Greater analysts' forecast accuracy is expected when a greater number of analysts are following the firm (Lys and Sohn, 1990). In addition, we will control for forecast dispersion (*DISP*), which is the standard deviation of the analysts' earnings forecasts, divided by the stock price at the end of year t-1. Greater forecast dispersion could indicate greater complexity in forecasting earnings, or more private information gathering (Barron, Kim, Lim, and Stevens, 1998). Lastly, we will also include the number of management earnings forecasts issued by the firm during the current fiscal year (*MFNUM*) to control for publicly available information about the firm.

We add analyst-level control variables. We control for the analysts' firm-specific experience by adding *A_EXP*, which is the natural logarithm of the number of years that an analyst has issued at least one EPS forecast for a given firm. More experienced analysts are likely to issue accurate forecasts because their forecasting skills and firm-specific knowledge improve over time (Clement, 1999). To control for the analysts' brokerage house size, we include *B_ANA*, which is the number of analysts at the brokerage house. Forecasting likely enhances with more resources available to analysts. To control for portfolio complexity, we include the number of industries (2-digit SIC code) that the analyst follows (*NoSIC2*) and the number of firms (*NoFirms*). Analysts could find it more difficult to follow more firms and industries (Clement, 1999). However, the

information spillover effect could improve analysts' forecasts (Clement et al., 2003; Bolliger, 2004). We include year, industry, and analyst fixed effects to control for any time-series, cross-analysts, and cross-industry variation.

Testing H2

To examine whether analysts' forecast accuracy is lower when multi-unit firms have a greater proportion of establishments in the early life-cycle stage, we will use the following two models:

$$ACCURACY_{ijt} = \beta_0 + \beta_1 EARLYPROP_{it} + \beta_4 SIZE_{it} + \beta_5 EVOL_{it} + \beta_6 SKEW_{it} + \beta_7 LOSS_{it} + \beta_8 BTM_{it} + \beta_9 DISP_{it} + \beta_{10} FOLLOW_{it} + \beta_{11} \Delta EARN_{it} + \beta_{12} MFNUM_{it} + \beta_{13} ESTABNUM_{it} + \beta_{14} A_EXP_{ijt} + \beta_{15} B_ANA_{ijt} + \beta_{16} NoSIC2_{ijt} + \beta_{17} NoFirms_{ijt} + \varepsilon_{ijt} \quad (3)$$

$$ACCURACY_{ijt} = \beta_0 + \beta_1 STG1_PROP_{it} + \beta_2 STG2_PROP_{it} + \beta_3 STG3_PROP_{it} + \beta_4 STG4_PROP_{it} + \beta_5 SIZE_{it} + \beta_6 EVOL_{it} + \beta_7 SKEW_{it} + \beta_8 LOSS_{it} + \beta_9 BTM_{it} + \beta_{10} DISP_{it} + \beta_{11} FOLLOW_{it} + \beta_{12} \Delta EARN_{it} + \beta_{13} MFNUM_{it} + \beta_{14} ESTABNUM_{it} + \beta_{15} A_EXP_{ijt} + \beta_{16} B_ANA_{ijt} + \beta_{17} NoSIC2_{ijt} + \beta_{18} NoFirms_{ijt} + \varepsilon_{ijt} \quad (4)$$

The dependent variable is analysts' forecast accuracy ($ACCURACY_{ijt}$). In Model (3),[8] *EARLY_PROP* is the proportion of the introduction and the growth stage establishments in a firm. For *EARLY_PROP*, we use *EARLY_TVS*, *EARLY_CAPEX*, *EARLY_EMP*, and *EARLY_PRODHR*.[9] *EARLY_TVS* (*EARLY_CAPEX*, *EARLY_EMP*, and *EARLY_PRODHR*) is the total introduction and growth stage establishments' *TVS* (*CAPEX*, *EMP*, and *PRODHR*) divided by the total *TVS* (*CAPEX*, *EMP*, and *PRODHR*) of the firm. For H2, we expect to find a significantly negative β_1 if analysts' forecast accuracy is lower for firms that have a greater proportion of establishments in the early life-cycle stages.

We also use the proportion of establishments in each life-cycle stage in Model (4). For

[8] We do not interact *PROP* and *ESTABNUM* because of the high multicollinearity caused by the interaction term.
[9] We also calculate *PROP* using the proportion based on *WAGE*. We get similar and significant results (untabulated).

calculating the establishment proportion of each stage, we use *TVS*, *CAPEX*, *EMP*, and *PRODHR*.[10] Specifically, *STGn_TVS* (*STGn_TCE*, *STGn_CAPEX*, and *STGn_PRODHR*) is the total *TVS* (*CAPEX*, *EMP*, and *PRODHR*) of stage *n* establishments divided by the total *TVS* (*CAPEX*, *EMP*, and *PRODHR*) of the firm where stage 1= introduction, stage 2= growth, stage 3= maturity, stage 4= shake-out, and stage 5= decline stage. We do not include stage 5 in the model as it is the base group. We expect to find a significantly negative β_1 and β_2 if analysts' forecast accuracy is lower for firms with a greater proportion of establishments in the introduction and growth stages.

4. Empirical results

Descriptive statistics

Panel A of Table 1 presents the descriptive statistics of the entire sample. Instead of the actual percentile, we use the pseudo percentile following the Census Bureau's policy.[11] To calculate the pseudo percentile, we calculate the average from at least 11 observations around the percentile where the subset of observations is comprised of at least five observations on either side.

The mean *SAMELC5* is 0.068, respectively, indicating that about 7 percent of multi-unit firms have all their establishments in the same life-cycle stage. The mean *EARLY_TVS* is 0.560, respectively, indicating that 56 percent of multi-unit firms' total value of shipments are generated from the introduction and the growth stage establishments. The mean values of *EARLY_CAPEX*, *EARLY_EMP*, and *EARLY_PRODHR* are similar to the mean *EARLY_TVS*.

While the mean *ESTABNUM* is 2.203, the mean *SEGMENTNUM* is 1.354, indicating that

[10] We also calculate proportions using the number of stage *n* establishments divided by the total number of establishments of a firm. We get similar and significant results using this variable (untabulated).

[11] The purpose of using the pseudo percentile is to reduce the disclosure risk and to protect the census data. The Census Bureau projects usually use the pseudo percentile rather than the true percentile including medians.

the number of establishments are greater than the reported segments in 10-Ks. The mean *SEG_SAMELC* is 0.203, respectively, indicating that about 20 percent of multi-unit firms have all their reported segments in the same life-cycle stage. These values are smaller than the mean *SAMELC* that are around 0.07. This difference suggests that firms' discretion in disclosing segments could lead to different life-cycle stages.

Panel B of Table 1 presents the mean and the standard deviation of two subsamples, *SAMELC*=1 and *SAMELC*=0. *ACCURACY* is higher in *SAMELC5*=1 than *SAMELC5*=0, suggesting that analysts' forecasts are more accurate when all the establishments of a multi-unit firm are in the same life-cycle stage than when the establishments are in different life-cycle stages.

Table 2 shows the Pearson correlation of the five variables (*TVS*, *CAPEX*, *EMP*, *PRODHR*, and *WAGE*) from the CMF. We find that all five variables are highly correlated, suggesting that calculating the proportion of establishments based on these variables leads to similar results for the second hypothesis. Therefore, we do not show all the results but show the proportion results in Tables 4, 5, 8, and 9 using the first four variables.

The relation between analysts' forecast accuracy and establishments' same life-cycle stages

Table 3 presents the OLS estimation results of Model (2). *SAMELC* has a significantly positive coefficient, suggesting that analysts' forecasts are more accurate when multi-unit firms have all their establishments in the same life-cycle stage. The coefficient on *ESTABNUM* is significantly negative, indicating that analysts' forecasts are less accurate when multi-unit firms have more establishments. This result suggests that analysts find it more difficult to forecast when there are more establishments to analyze due to their limited resources and time. *ESTABNUM*SAMELC* has a significantly negative coefficient, indicating that analysts' forecasts

are more accurate when multi-unit firms have fewer establishments and when all the establishments are in the same life-cycle stage. These results support H1.

Overall, we find that it is relatively easier for analysts to forecast multi-unit firms when all their establishments are in the same life-cycle stage than when the firms' establishments are in more than one life-cycle stage. These findings suggest that analysts' forecasts could become more accurate if they could save their effort in analyzing different signals in different life-cycle stages and reduce their work assigning different weights to the signals depending on the establishments' life-cycle stage.

The relation between analysts' forecast accuracy and establishments' early life-cycle stages

Table 4 presents the OLS estimation results of Model (3). For *EARLYPROP*, we use *EARLY_TVS*, *EARLY_CAPEX*, *EARLY_EMP*, and *EARLY_PRODHR* in columns (1), (2), (3), and (4), respectively. The estimated coefficients on *EARLYPROP* are negative and significant at the 1% level, indicating that analysts' forecasts are less accurate when the proportion of establishments in the early life-cycle stages (i.e., the introduction stage and the growth stage) is high. For *EARLYPROP*, we also measure the proportion based on the number of establishments (i.e., the number of early-stage establishments divided by the total number of establishments). The results are similar and significant (untabulated). These results suggest that it is more difficult to forecast firms that have a greater proportion of early-stage establishments due to greater uncertainty in the early life-cycle stages.

Table 5 shows the OLS estimation results of Model (4). For *STGn_PROP*, we use *STGn_*TVS, *STGn_CAPEX*, *STGn_EMP*, and *STGn_PRODHR* in columns (1), (2), (3), and (4), respectively. *STG1_PROP* is significantly negative in all the columns, indicating that analysts'

forecasts are less accurate whenmulti-unit firms have a higher proportion of establishments in the introduction stage. *STG2_PROP* is significantly negative in all columns, indicating that analysts' forecasts are less accurate when firms have a greater proportion of establishments in the growth stage. These results are consistent with the results in Table 4 where we find that firms with more early-stage establishments are associated with less accurate analysts' forecasts.

Overall, our findings suggest that firms with more mature and shake-out stages are relatively easier to forecasts earnings compared to the firms with more introduction and growth stages possibly because earnings in the mature and shake-out stages are less uncertain than earnings in the earlier stages.

5. Additional analysis

Limitations of the segment classification based on 10-K reports

This study estimates the life cycle for each unit within a firm using the establishment classification from the Census data. We argue that using the establishment classification is imperative because of the limitations in the publicly available data. The unit information can be obtained from firms' annual filings (i.e., the 10-K filings) with the SEC. The Financial Accounting Standards Board (FASB) mandates that public firms report information about operating segments in their financial statements (FASB Accounting Standards Codification Topic 280, *Segment Reporting*). ASC 280 also requires that firms disclose information about their products and services, the geographic areas in which they operate, and their major customers. Firms can determine what information to report based on how they organize the segments for making operating decisions and assessing performance.

Because firms have discretion in disclosing their segments, reported segments have several

data problems. First, firms tend to disclose fewer segments. For instance, a firm can report only one segment in its 10-K reports even if it has multiple business units. Fewer segments in the 10-K filings are allowed by the criteria in the standards that let firms aggregate units with similar economic characteristics. However, firms' 10-K segment classifications may also result from firms' strategic aggregation to avoid disclosing proprietary data. Second, it is challenging for investors or other information users to compare segments across firms because firms could define segments differently. Some firms classify their segments primarily based on their products and services, whereas other firms primarily classify their segments based on geographic locations. Third, it is difficult to compare the same segment's performance across time because a firm's definition of a segment could change. Thus, although a firm is disclosing the "same" segment over time, the establishments included in the segment likely change. Therefore, we believe that the establishment classification using the Census Bureau data would improve the estimation of the life cycle.

Segment-level life-cycle estimation based on 10-K reports

For comparison purposes, we estimate the life cycle of 10-K segments using the same method employed in estimating the establishment-level life cycles. For the life-cycle descriptors, we use segment sales, segment capex, and segment employment. We choose these three variables from the 10-K segment reporting because the three variables overlap with our establishment-level life-cycle descriptors from the Census Bureau data. We do not use segment age because segments often change and merge with other segments over time, leading to less accurate age calculation.

In Panel A of Table 6, we provide the Pearson correlation between the proportions of firms' segments in five life-cycle stages using the 10-K segment sales (*SEG_STGn_SALES*) and the proportions of firms' establishments in the five stages using the Census Bureau's establishment

TVS (*STGn_TVS*). In Panel B, we show the correlation between the proportions using the 10-K segment capex (*SEG_STGn_CAPEX*) and the proportions using the establishment *CAPEX* (*STGn_CAPEX*). In Panel C, we show the correlation between the proportions using the 10-K segment employment (*SEG_STGn_EMP*) and the proportions using the establishment *EMP* (*STGn_EMP*).

In Panel A, the correlation coefficients of *STG1* (introduction stage), *STG4* (shake-out stage), and *STG5* (decline stage) variables are positive while the *STG2* (growth stage) coefficient is negative. In Panel B, the correlation coefficients on the diagonal line from the top left to the bottom right are positive except for the *STG3* (mature stage) coefficient. These correlation coefficients on the diagonal line suggest that life cycle estimations based on segment classification and based on establishment classification overlap but the magnitude of the correlation is not very high. In Panel C, only *STG1* has a positive coefficient and the size of the coefficient is smaller than in the other two panels. It could be because segment employment data have more missing values than sales and capex data.

The relation between analysts' forecast accuracy and establishments' same life-cycle stages using 10-K segments for life-cycle estimation

We examine whether our H1 findings still hold when we use an alternative life-cycle estimation based on 10-K segments. In Panel A of Table 7, we estimate Model (2) replacing *SAMELC* with *SEG_SAMELC*. We use the natural logarithm of the number of segments disclosed in 10-K reports (i.e., *SEGMENTNUM*) instead of *ESTABNUM*. We find significantly positive coefficients on *SEGMENTNUM*, indicating that analysts' forecasts are more accurate when firms disclose more segments. The sign of the coefficient is the opposite of the coefficient on

ESTABNUM in Table 4. This result suggests that having more disclosed segments are better for analysts' forecasting because segment disclosures provide useful information to analysts for forecasting. Since some firms try to hide their proprietary information by aggregating multiple operations, having detailed information about more segments could improve analysts' forecasts.

Overall, the results allude to the discretionary nature of segment reporting and the valuable information embedded in the 10-K segment information. The establishment-level census data, however, are only reported to the Census Bureau and are not disclosed publicly. Analysts should research and investigate more when the number of establishments increases, especially when the establishments are in different life-cycle stages, making it more difficult for analysts to forecast earnings.

Next, we examine whether our H2 results hold when we use 10-K segments for life-cycle estimation. In Panel B of Table 7, we show the estimation results of Model (4) using the segment proportions in each life cycle stage (i.e., *SEG_STGn_PROP*). We use segment sales, capex, and employment to calculate the proportions. That is, we use the total segment sales (capex, and employment) of stage *n* segments divided by the total segment sales (capex, and employment) of the firm where stage 1= introduction, stage 2= growth, stage 3= maturity, stage 4= shake-out, and stage 5= decline stage. The proportions calculated using segment sales, capex, and employment are shown in columns (1), (2), and (3), respectively.

We find that *SEG_STG1_PROP* and *SEG_STG2_PROP* have significantly negative coefficients in all columns, indicating that analysts' forecasts are less accurate when the proportions of 10-K segments in the introduction stage and the growth stage are high. The coefficient for *SEG_STG3_PROP* using segment employment is significantly positive in column (3), suggesting that analysts' forecasts are more accurate when more segments are in the mature

stage. However, *SEG_STG4_PROP* using segment sales is significantly negative in column (1), suggesting that analysts' forecasts are less accurate when more segments are in the shake-out stage. Overall, these results are consistent with our results using the establishment proportions in each life cycle stage.

The gap between the number of establishments and the number of 10-K segments

We further investigate how the gap between the number of establishments and the number of 10-K segments of a multi-unit firm affects analysts' forecast accuracy. We define *GAP* as the difference between the number of establishments and the number of 10-K segments. In Table 8, we present the estimation results using *GAP*, the establishment proportions in each life-cycle stage (i.e., *STGn_PROP* where *n*=1, 2, 3, and 4), and the interactions between *GAP* and the proportions. In column (1) where we examine the effect of *GAP* alone, *GAP* has a significantly negative coefficient, indicating that analysts' forecasts are less accurate when firms disclose fewer segments than the actual number of establishments. It suggests that it is more difficult to forecast multi-unit firms' earnings when firms aggregate or hide more establishments in their segment reporting.

The association between management forecast accuracy and the proportion of establishments in different life-cycle stages

In addition to examining the analysts' forecast accuracy, we examine how management forecast accuracy is related to the proportion of multi-unit firms' establishments in different life-cycle stages. While analysts find it more difficult to forecast firms with more establishments in the early life-cycle stages, managers might find it easier to forecast future earnings since they have better information about their establishments. On the other hand, managers could also find it challenging to forecast earnings when more establishments are in the early-stage life cycle stages.

The greater uncertainty in the early stages could complicate not only analysts' forecasting but also managers' forecasting of earnings.

Using the management guidance data from the I/B/E/S Guidance, we estimate Model (4) with management forecast accuracy (*MFACCURACY*) for the dependent variable and show the results in Table 9. We define *MFACCURACY* as (-1)*|management EPS forecast - actual EPS| divided by the price at the end of the previous fiscal year-end. Just like Table 5, the proportions of establishments in each stage are calculated using *TVS* in column (1), *CAPEX* in column (2), *EMP* in column (3), and *PRODHR* in column (4). We find significantly negative coefficients on *STG1_PROP* for all the columns, indicating that managers' earnings forecast accuracy is lower when firms have a greater proportion of establishments in the introduction stage. *STG2_PROP* has significantly negative coefficients in columns (1) and (4), indicating that managers' forecasts are less accurate when firms have a greater proportion of establishments in the growth stage. These results suggest that managers also find it difficult to forecast earnings for multi-unit firms that have a high proportion of early-stage establishments.

6. Conclusion

This paper examines how multi-unit firms' establishment-level life-cycle stages influence analysts' forecast accuracy. Instead of using the firm-level life cycle, we focus on the establishment-level life cycle because estimating the life cycle at the firm level could result in inaccurate life-cycle estimation especially when different establishments are in different life-cycle stages. We use six life-cycle descriptors from the Census Bureau's establishment data to estimate the establishment-level life cycle.

We find that analyst forecasts are less accurate for multi-unit firms whose establishments

are in different life-cycle stages than those in the same life-cycle stage. This finding suggests that the forecasting difficulty of more diversified firms can be attributed to the different life-cycle stages of each establishment. When all the establishments are in the same life-cycle stage, analysts could save their time and effort in analyzing the signals in different life-cycle stages. We also find that analysts' forecast accuracy is lower for firms that have a greater proportion of establishments in the early life-cycle stage, suggesting that the introduction and the growth stages are associated with greater uncertainty.

By using the 10-K segments rather than the establishments for the life cycle estimation, we find that analysts' forecast accuracy is higher when firms have more disclosed segments. We also find that analysts' forecast accuracy is lower when firms report fewer segments than the number of establishments. These findings suggest that the 10-K segment disclosures provide useful information but aggregating more establishments for segment reporting could complicate analysts' forecasting.

Overall, the findings of this paper suggest that the establishment-level life cycle could help us better understand the actual life cycle of firms' subunits. We argue that analysts and investors should be more cautious with the firm-level life cycle as the aggregation of multiple life cycles could lead to an inaccurate life cycle analysis. Moreover, when we only consider the firm-level life cycle, we tend to ignore the various signals from multiple subunits' various life cycle stages that have different implications for future earnings. Because establishment-level life cycles are related to establishments' evolution along an expected path, forecasts that incorporates these life cycles will be more valuable for less sophisticated investors. Analysts' understanding of different life cycles within a firm will help investors more accurately value a firm.

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APPENDIX

Variable definition

Variable	Definition
ACCURACY	(-1)*|analyst's earnings-per-share (eps) forecast in year t minus the actual eps| divided by the price at the end of the previous fiscal year-end.
SAMELC	A dummy variable that is equal to one if all of the firm's establishments are in the same life cycle stage from the 5-stage life cycle (using rankings of all establishments) and zero otherwise.
EARLYPROP	Alternative variables are used for *EARLYPROP*: *EARLY_TVS*, *EARLY_CAPEX*, *EARLY_EMP*, and *EARLY_PRODHR*, (which are defined below).
EARLY_TVS5	Total early-stage establishments' TVS (total value of shipments) divided by the total TVS of a firm, where the life-cycle stage is estimated based on the life-cycle descriptors' rankings for all establishments.
EARLY_CAPEX5	Total early-stage establishments' capex divided by the total capex of a firm, where the life-cycle stage is estimated based on the life-cycle descriptors' rankings for all establishments.
EARLY_EMP5	Total early-stage establishments' total employment divided by the total employment of a firm, where the life-cycle stage is estimated based on the life-cycle descriptors' rankings for all establishments.
EARLY_PRODHR5	Total early-stage establishments' total production worker hours divided by the total production worker hours of a firm, where life-cycle stage is estimated based on the life-cycle descriptors' rankings for all establishments.
STGn_PROP (where n = 1, 2, 3, 4, and 5)	Alternative variables are used for *STGn_PROP* (where *n*= 1, 2, 3, 4, and 5): *STGn_TVS*, *STGn_CAPEX*, *STGn_EMP*, and *STGn_PRODHR* (which are defined below).
STGn_TVS	The total TVS (total value of shipments) of stage *n* establishments divided by the total TVS of establishments of a firm, where the life-cycle stage is estimated based on the life-cycle descriptors' rankings for all establishments.
STGn_CAPEX	The total CAPEX (capex) of stage *n* establishments divided by the total CAPEX of establishments of a firm, where the life-cycle stage is estimated based on the life-cycle descriptors' rankings for all establishments.
STGn_EMP	The total EMP (employment) of stage *n* establishments divided by the total EMP of establishments of a firm, where the life-cycle stage is estimated based on the life-cycle descriptors' rankings for all establishments.
STGn_PRODHOUR	The total PRODHR (production worker hours) of stage *n* establishments divided by the total PRODHR of establishments of a firm, where the life-cycle stage is estimated based on the life-cycle descriptors' rankings for all establishments.
SEG_SAMELC	A dummy variable that is equal to one if all of the firm's 10-K segments are in the same life cycle stage from the 5-stage life cycle (using rankings of all segments) and zero otherwise.

SEG_STGn_PROP (where n = 1, 2, 3, 4, and 5)	Alternative variables are used for *SEG_STGn_PROP* (where *n*= 1, 2, 3, 4, and 5): *STGn_sales*, *STGn_capx*, and *STGn_emp* (which are defined below).
STGn_sales	The total sales of stage *n* 10-K segments divided by the total sales of 10-K segments of a firm, where the life-cycle stage is estimated based on the life-cycle descriptors' rankings for all 10-K segments.
STGn_capx	The total capex of stage *n* 10-K segments divided by the total capex of 10-K segments of a firm, where the life-cycle stage is estimated based on the life-cycle descriptors' rankings for all 10-K segments.
STGn_emp	The total employment of stage *n* 10-K segments divided by the total employment of the 10-K segments of a firm, where the life-cycle stage is estimated based on the life-cycle descriptors' rankings for all 10-K segments.
SEGMENTNUM	The natural log of (1+ the number of the firm's 10-K segments in year t).
GAP	The absolute value of the difference between *ESTABNUM* and *SEGMENTNUM*.
MFACCURACY	(-1)*|management eps forecast in year t minus the actual eps| divided by the price at the end of the previous fiscal year-end
SIZE	The natural log of(total assets) in year t
EVOL	The standard deviation of return on assets over the previous five years
SKEW	The skewness of the past ten years of earnings per share
LOSS	An indicator variable that equals one if the firm's net income is negative in year t
BTM	The book value of equity divided by the market value of equity in year t
DISP	The standard deviation of analysts' annual earnings forecast for the given firm in year t
FOLLOW	The natural log of (1+the number of analysts following) the firm in year t
Δ*EARN*	The absolute value of the difference between the current year's eps and the last year's eps, divided by the price at the end of the previous year.
MFNUM	The natural log of (1+ the number of management's annual eps forecasts issued by the firm in year t
ESTABNUM	The natural log of (1+ the number of the firm's establishments in year t)
A_EXP	The natural log of (1+ the number of years that an analyst has issued at least one earnings forecast for a given firm)
B_ANA	The natural log of (1+ the number of analysts at the brokerage house)
NoSIC2	The natural log of (1+ the number of 2-digit SIC code industries that the analyst follows)
NoFirms	The natural log of (1+ the number of firms that the analyst follows)

TABLE 1
Descriptive statistics

Panel A. Descriptive statistics for the full sample

VARIABLES	N	Mean	SD	Pseudo P25	Pseudo P50	Pseudo P75
ACCURACY	63,500	-0.057	0.108	-0.062	-0.028	-0.009
SAMELC	63,500	0.068	0.252	0.000	0.000	0.000
EARLY_TVS	63,500	0.560	0.348	0.240	0.611	0.884
EARLY_CAPEX	63,500	0.579	0.360	0.257	0.661	0.933
EARLY_EMP	63,500	0.551	0.338	0.249	0.595	0.844
EARLY_PRODHR	63,500	0.562	0.338	0.289	0.620	0.864
SIZE	63,500	7.365	1.650	6.219	7.321	8.443
EVOL	63,500	0.046	0.039	0.020	0.033	0.059
SKEW	63,500	-0.063	0.948	-0.623	0.063	0.548
LOSS	63,500	0.126	0.332	0.000	0.000	0.000
BTM	63,500	0.544	0.332	0.304	0.498	0.724
DISP	63,500	0.013	0.033	0.002	0.004	0.011
FOLLOW	63,500	2.724	0.791	2.303	2.890	3.332
Δ*EARN*	63,500	0.052	0.090	0.009	0.021	0.051
MFNUM	63,500	0.178	0.509	0.000	0.000	0.000
ESTABNUM	63,500	2.203	0.948	1.386	2.079	2.773
A_EXP	63,500	1.471	0.561	1.099	1.386	1.946
B_ANA	63,500	2.643	1.084	1.946	2.639	3.466
NoSIC2	63,500	1.704	0.537	1.386	1.792	2.079
NoFirms	63,500	2.777	0.532	2.485	2.773	3.135
SEG_SAMELC	22,500	0.203	0.402	0.000	0.000	0.000
SEGMENTNUM	22,500	1.354	0.264	1.099	1.386	1.792
GAP	22,500	1.270	0.941	0.511	1.099	1.969
MFACCURACY	650	-0.024	0.039	-0.026	-0.012	-0.004

Panel B. Descriptive statistics for *SAMELC*=1 and *SAMELC*=0

	***SAMELC5*=1**			***SAMELC5*=0**			
VARIABLES	N	Mean	SD	N	Mean	SD	
ACCURACY	4,300	-0.048	0.083	59,200	-0.058	0.110	***
SIZE	4,300	6.816	1.773	59,200	7.406	1.633	***
EVOL	4,300	0.060	0.040	59,200	0.045	0.038	***
SKEW	4,300	-0.433	1.146	59,200	-0.036	0.927	***
LOSS	4,300	0.141	0.348	59,200	0.125	0.331	***
BTM	4,300	0.388	0.324	59,200	0.555	0.330	***
DISP	4,300	0.010	0.031	59,200	0.014	0.033	***
FOLLOW	4,300	2.505	0.901	59,200	2.741	0.780	***
Δ*EARN*	4,300	0.041	0.073	59,200	0.053	0.092	***
MFNUM	4,300	0.254	0.605	59,200	0.172	0.501	***
ESTABNUM	4,300	1.284	0.254	59,200	2.271	0.945	***
A_EXP	4,300	1.523	0.560	59,200	1.467	0.561	***
B_ANA	4,300	2.645	1.104	59,200	2.643	1.083	
NoSIC2	4,300	1.637	0.552	59,200	1.709	0.536	***
NoFirms	4,300	2.765	0.471	59,200	2.778	0.536	

These tables show the descriptive statistics. Panel A presents the summary statistics for the entire sample. Panel B shows the mean and standard deviation for two subsamples. It shows the descriptive statistics for *SAMELC5*=1 and *SAMELC5*=0, respectively. The last column shows the significance of the *t*-test of the two subsamples. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

TABLE 2

Correlation of *TVS*, *CAPEX*, *EMP*, *WAGE*, *TAE*, and *PRODHR*

	TVS	***CAPEX***	***EMP***	***WAGE***	***PRODHR***
TVS	1				
CAPEX	0.629***	1			
EMP	0.780***	0.372***	1		
WAGE	0.920***	0.374***	0.898***	1	
PRODHR	0.776***	0.422***	0.924***	0.806***	1

This table presents the Pearson correlation of the five variables: the total value of shipment (*TVS*), the total capex (*CAPEX*), the total employment (*EMP*), the total production worker hours (*PRODHR*), and the total salary and wages (*WAGE*) of an establishment. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

TABLE 3

Analysts' forecast accuracy and firms' establishments in the same life-cycle stage

	(1)
VARIABLES	
SAMELC	0.054***
	(3.50)
ESTABNUM	-0.004***
	(-2.73)
*ESTABNUM*SAMELC*	-0.045***
	(-4.12)
SIZE	-0.003*
	(-1.77)
EVOL	-0.096**
	(-1.98)
SKEW	-0.003**
	(-2.01)
LOSS	-0.031***
	(-5.58)
BTM	0.035***
	(6.50)
DISP	-2.195***
	(-16.58)
FOLLOW	0.009***
	(3.15)
Δ*EARN*	-0.097***
	(-4.28)
MFNUM	0.004
	(1.48)
A_EXP	0
	(-0.13)
B_ANA	0.003
	(1.42)
NoSIC2	-0.003
	(-0.68)
NoFirms	-0.002
	(-0.75)
Constant	-0.024*
	(-1.85)
Adj R^2	0.631
Year/Analyst/Industry FE	yes
Observations	63,500

This table presents the OLS estimation results of the following model:

$$ACCURACY_{ijt} = \beta_0 + \beta_1 SAMELC_{it} + \beta_2 ESTABNUM_{it} + \beta_3 SAMELC * ESTABNUM_{it} + \beta_4 SIZE_{it} + \beta_5 EVOL_{it} + \beta_6 SKEW_{it} + \beta_7 LOSS_{it} + \beta_8 BTM_{it} + \beta_9 DISP + \beta_{10} FOLLOW_{it} + \beta_{11} \Delta EARN_{it} + \beta_{12} MFNUM_{it} + \beta_{13} A_EXP_{ijt} + \beta_{14} B_ANA_{ijt} + \beta_{15} NoSIC2_{ijt} + \beta_{16} NoFirms_{ijt} + \varepsilon_{ijt}$$

where *ACCURACY* is (-1) multiplied by the absolute value of the analyst's eps forecast in year t minus the actual eps, divided by the price at the end of the previous fiscal year-end. *SAMELC* is a dummy variable that is equal to one if all of the firm's establishments are in the same life cycle stage from the 5-stage life cycle and zero otherwise. *ESTABNUM* is the natural log of (1+ the number of the firm's establishments in year t). Other variables are defined in Appendix. *T*-statistics are shown in the parenthesis. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

TABLE 4

Analysts' forecast accuracy and the proportion of establishments in the early life-cycle stages

	(1)	(2)	(3)	(4)
VARIABLES	*TVS*	*CAPEX*	*EMP*	*PRODHR*
EARLYPROP	-0.016***	-0.024***	-0.019***	-0.018***
	(-5.12)	(-7.93)	(-6.24)	(-5.61)
SIZE	-0.003**	-0.004**	-0.003**	-0.003**
	(-2.18)	(-2.45)	(-2.12)	(-2.25)
EVOL	-0.093*	-0.100**	-0.090*	-0.095**
	(-1.94)	(-2.07)	(-1.88)	(-1.98)
SKEW	-0.003*	-0.003*	-0.003*	-0.003*
	(-1.85)	(-1.82)	(-1.81)	(-1.85)
LOSS	-0.033***	-0.034***	-0.033***	-0.033***
	(-5.77)	(-5.94)	(-5.88)	(-5.82)
BTM	0.033***	0.030***	0.032***	0.033***
	(6.30)	(5.81)	(6.10)	(6.22)
DISP	-2.191***	-2.185***	-2.186***	-2.186***
	(-16.59)	(-16.66)	(-16.59)	(-16.60)
FOLLOW	0.009***	0.009***	0.009***	0.009***
	(3.41)	(3.41)	(3.38)	(3.40)
Δ*EARN*	-0.103***	-0.106***	-0.104***	-0.104***
	(-4.53)	(-4.71)	(-4.62)	(-4.61)
MFNUM	0.003	0.003	0.003	0.003
	(1.35)	(1.19)	(1.12)	(1.12)
ESTABNUM	-0.004***	-0.004**	-0.004***	-0.004**
	(-2.63)	(-2.29)	(-2.70)	(-2.50)
A_EXP	-0.001	-0.001	-0.001	-0.001
	(-0.31)	(-0.45)	(-0.41)	(-0.44)
B_ANA	0.003	0.003	0.003	0.003
	(1.37)	(1.38)	(1.34)	(1.34)
NoSIC2	-0.002	-0.002	-0.003	-0.003
	(-0.52)	(-0.57)	(-0.61)	(-0.64)
NoFirms	-0.003	-0.002	-0.002	-0.002
	(-0.85)	(-0.79)	(-0.75)	(-0.73)
Constant	-0.011	-0.002	-0.009	-0.009
	(-0.92)	(-0.19)	(-0.71)	(-0.69)
Adj R^2	0.632	0.634	0.632	0.632
Year/Analyst/Ind FE	yes	yes	yes	yes
Observations	63,500	63,500	63,500	63,500

This table presents the OLS estimation results of the following model:

$$ACCURACY_{ijt} = \beta_0 + \beta_1 PROP_{it} + \beta_4 SIZE_{it} + \beta_5 EVOL_{it} + \beta_6 SKEW_{it} + \beta_7 LOSS_{it} + \beta_8 BTM_{it} + \beta_9 DISP_{it} + \beta_{10} FOLLOW_{it} + \beta_{11} \Delta EARN_{it}$$

$$+\beta_{12}MFNUM_{it} + \beta_{13}ESTABNUM_{it} + \beta_{14}A_EXP_{ijt} + \beta_{15}B_ANA_{ijt} + \beta_{16}NoSIC2_{ijt} + \beta_{17}NoFirms_{ijt} + \varepsilon_{ijt}$$

where *ACCURACY* is (-1) multiplied by the absolute value of the analyst's eps forecast in year t minus the actual eps, divided by the price at the end of the previous fiscal year-end. For *EARLYPROP*, we use the following variables. For *EARLYPROP*, we use *EARLY_TVS*, *EARLY_CAPEX*, *EARLY_EMP*, and *EARLY_PRODHR* in columns (1), (2), (3), and (4), respectively. *EARLY_TVS* (*EARLY_CAPEX*; *EARLY_EMP*; *EARLY_PRODHR*) is the total *TVS* (*CAPEX*; *EMP*; *PRODHR*) of the early-stage (i.e., stages 1 and 2) establishments divided by the total *TVS* (*CAPEX*; *EMP*; *PRODHR*) of establishments of a firm, where the life-cycle stage is estimated based on the life-cycle descriptors' rankings for all establishments. *TVS* is the total value of shipments, *CAPEX* is the total capex, *EMP* is the total employment, and *PRODHR* is the total production worker hours of an establishment. Other variables are defined in Appendix. *T*-statistics are shown in the parenthesis. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

TABLE 5

Analysts' forecast accuracy and the proportion of establishments in each life-cycle stage

VARIABLES	(1) *TVS*	(2) *CAPEX*	(3) *EMP*	(4) *PRODHR*
STG1_PROP	-0.020***	-0.018***	-0.024***	-0.015**
	(-3.34)	(-3.10)	(-3.56)	(-2.16)
STG2_PROP	-0.015**	-0.023***	-0.024***	-0.024***
	(-2.38)	(-3.49)	(-3.14)	(-3.41)
STG3_PROP	0.001	0.008	-0.004	0.000
	(0.16)	(1.46)	(-0.54)	(0.03)
STG4_PROP	-0.005	-0.003	-0.008	-0.003
	(-0.70)	(-0.40)	(-1.08)	(-0.41)
SIZE	-0.003**	-0.004**	-0.003**	-0.003**
	(-2.19)	(-2.43)	(-2.11)	(-2.14)
EVOL	-0.090*	-0.098**	-0.089*	-0.097**
	(-1.88)	(-2.02)	(-1.86)	(-2.02)
SKEW	-0.003**	-0.003*	-0.003*	-0.003*
	(-1.96)	(-1.90)	(-1.94)	(-1.85)
LOSS	-0.033***	-0.033***	-0.033***	-0.033***
	(-5.81)	(-5.89)	(-5.86)	(-5.73)
BTM	0.033***	0.030***	0.032***	0.033***
	(6.13)	(5.57)	(6.01)	(6.23)
DISP	-2.195***	-2.187***	-2.190***	-2.183***
	(-16.65)	(-16.94)	(-16.73)	(-16.85)
FOLLOW	0.010***	0.009***	0.009***	0.009***
	(3.39)	(3.28)	(3.33)	(3.19)
Δ*EARN*	-0.102***	-0.106***	-0.104***	-0.104***
	(-4.53)	(-4.74)	(-4.62)	(-4.64)
MFNUM	0.003	0.003	0.003	0.003
	(1.34)	(1.01)	(1.16)	(1.04)
ESTABNUM	-0.004**	-0.003**	-0.004**	-0.004**
	(-2.35)	(-2.06)	(-2.54)	(-2.39)
A_EXP	-0.001	-0.001	-0.001	-0.001
	(-0.36)	(-0.37)	(-0.42)	(-0.34)
B_ANA	0.003	0.003	0.003	0.003
	(1.37)	(1.38)	(1.34)	(1.34)
NoSIC2	-0.002	-0.002	-0.002	-0.003
	(-0.44)	(-0.54)	(-0.56)	(-0.68)
NoFirms	-0.003	-0.003	-0.003	-0.002
	(-0.94)	(-0.85)	(-0.81)	(-0.71)
Constant	-0.01	-0.005	-0.004	-0.008
	(-0.77)	(-0.42)	(-0.33)	(-0.63)
Adj R^2	0.632	0.634	0.632	0.632

Year/Analyst/Ind FE	yes	yes	yes	yes
Observations	63,500	63,500	63,500	63,500

This table presents the OLS estimation results of the following model:

$$ACCURACY_{ijt} = \beta_0 + \beta_1 STG1_PROP_{it} + \beta_2 STG2_PROP_{it} + \beta_3 STG3_PROP_{it} + \beta_4 STG4_PROP_{it} + \beta_5 SIZE_{it} + \beta_6 EVOL_{it} + \beta_7 SKEW_{it} + \beta_8 LOSS_{it} + \beta_9 BTM_{it} + \beta_{10} DISP_{it} + \beta_{11} FOLLOW_{it} + \beta_{12} \Delta EARN_{it} + \beta_{13} MFNUM_{it} + \beta_{14} ESTABNUM_{it} + \beta_{15} A_EXP_{ijt} + \beta_{16} B_ANA_{ijt} + \beta_{17} NoSIC2_{ijt} + \beta_{18} NoFirms_{ijt} + \varepsilon_{ijt}$$

where *ACCURACY* is (-1) multiplied by the absolute value of the analyst's eps forecast in year t minus the actual eps, divided by the price at the end of the previous fiscal year-end. For *STGn_PROP*, we use *STGn_TVS*, *STGn_CAPEX*, *STGn_EMP*, and *STGn_PRODHR* in columns (1), (2), (3), and (4), respectively. *STGn_TVS* (*STGn_CAPEX*; *STGn_EMP*; *STGn_PRODHR*) is the total *TVS* (*CAPEX*; *EMP*; *PRODHR*) of stage *n* establishments divided by the total *TVS* (*CAPEX*; *EMP*; *PRODHR*) of establishments of a firm, where the life-cycle stage is estimated based on the life-cycle descriptors' rankings for all establishments. *TVS* is the total value of shipments, *CAPEX* is the total capex, *EMP* is the total employment, and *PRODHR* is the total production worker hours of an establishment. Other variables are defined in Appendix. *T*-statistics are shown in the parenthesis. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

TABLE 6

Correlation between the proportion of establishments and the proportional of segments in each stage

Panel A. *STGn_TVS* and *SEG_STGn_SALES*

	STG1_TVS	*STG2_TVS*	*STG3_TVS*	*STG4_TVS*	*STG5_TVS*
SEG_STG1_SALES	0.343***	-0.058***	-0.181***	-0.110***	-0.080***
SEG_STG2_SALES	0.306***	-0.042***	-0.230***	-0.057***	-0.033***
SEG_STG3_SALES	-0.169***	0.317***	-0.012	-0.156***	-0.058***
SEG_STG4_SALES	-0.220***	-0.128***	0.281***	0.117***	0.036***
SEG_STG5_SALES	-0.221***	-0.043***	0.086***	0.166***	0.121***

Panel B. *STGn_CAPEX* and *SEG_STGn_CAPEX*

	STG1_CAPEX	*STG2_CAPEX*	*STG3_CAPEX*	*STG4_CAPEX*	*STG5_CAPEX*
SEG_STG1_CAPEX	0.213***	0.0211**	-0.176***	-0.042***	-0.134***
SEG_STG2_CAPEX	0.392***	-0.132***	-0.219***	-0.048***	-0.049***
SEG_STG3_CAPEX	-0.149***	0.263***	0.062***	-0.200***	-0.110***
SEG_STG4_CAPEX	-0.283***	-0.060***	0.300***	0.063***	0.099***
SEG_STG5_CAPEX	-0.197***	0.013	0.060***	0.178***	0.035***

Panel C. *STGn_EMP* and *SEG_STGn_EMP*

	STG1_EMP	*STG2_EMP*	*STG3_EMP*	*STG4_EMP*	*STG5_EMP*
SEG_STG1_EMP	0.141***	-0.057***	-0.071***	-0.035***	0.011
SEG_STG2_EMP	0.031***	-0.036***	0.063***	-0.036***	-0.042***
SEG_STG3_EMP	0.002	0.084***	-0.094***	-0.029***	0.039***
SEG_STG4_EMP	0.035***	0.039***	-0.056***	-0.008	-0.033***
SEG_STG5_EMP	0.016*	-0.029***	0.053***	-0.017*	-0.039***

These tables present the correlation between establishment proportions and segment proportions in the life-cycle stages measured using the Census Bureau's establishment data and firms' 10-K segment data, respectively. Panel A shows the Pearson correlation between the proportions of firms' segments in five life-cycle stages using the 10-K segment sales (*SEG_STGn_SALES*) and the proportions of firms' establishments in the five stages using the establishment *TVS* (*STGn_TVS*) where *n*= 1, 2, 3, 4, and 5. Panel B shows the correlation between the proportions using the 10-K segment capex (*SEG_STGn_CAPEX*) and the proportions using the establishment *CAPEX* (*STGn_CAPEX*). Panel C shows the correlation between the proportions using the 10-K segment employment (*SEG_STGn_EMP*) and the proportions using the establishment *EMP* (*STGn_EMP*). ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

TABLE 7

Alternative life-cycle estimation using 10-K segments

Panel A. Analysts' forecast accuracy and firms' 10-K segments in the same life-cycle stage

	(1)
VARIABLES	
SEG_SAMELC	0.043
	(0.94)
SEGMENTNUM	0.039***
	(7.14)
*SEGMENTNUM*SEG_SAMELC*	-0.036
	(-0.91)
LOSS	-0.078***
	(-7.01)
SIZE	-0.004**
	(-2.15)
EVOL	0.096
	(1.59)
SKEW	0.006***
	(3.67)
BTM	0.038***
	(5.94)
DISP	-1.509***
	(-5.17)
FOLLOW	0.010
	(1.48)
Δ*EARN*	-0.257***
	(-11.02)
MFNUM	0.006
	(1.29)
A_EXP	0.006
	(1.41)
B_ANA	0.004
	(1.12)
NoSIC2	0.001
	(0.18)
NoFirms	0.000
	(0.08)
Constant	-0.106***
	(-4.49)
Adj R^2	0.622
Year/Analyst/Ind FE	yes
Observations	22,500

Panel B. Analysts' forecast accuracy and the proportion of 10-K segments in each life-cycle stage

	(1)	(2)	(3)
VARIABLES	Segment Sales	Segment Capex	Segment Employment
SEG_STG1_PROP	-0.051***	-0.024**	-0.292***
	(-6.37)	(-2.11)	(-2.66)
SEG_STG2_PROP	-0.047***	-0.020***	-0.108***
	(-6.41)	(-2.78)	(-4.94)
SEG_STG3_PROP	-0.002	0.013	0.068***
	(-0.29)	(1.57)	(4.69)
SEG_STG4_PROP	-0.015***	0.004	0.133
	(-4.13)	(0.72)	(1.18)
LOSS	-0.083***	-0.079***	-0.080***
	(-7.46)	(-7.07)	(-7.24)
SIZE	0.001	0.000	0.001
	(0.28)	(-0.12)	(0.30)
EVOL	0.004	0.021	0.135**
	(0.07)	(0.35)	(2.43)
SKEW	0.005***	0.007***	0.007***
	(3.57)	(4.13)	(4.30)
BTM	0.033***	0.033***	0.023***
	(5.52)	(5.58)	(4.49)
DISP	-1.527***	-1.510***	-1.502***
	(-5.64)	(-5.27)	(-5.33)
FOLLOW	0.004	0.005	0.003
	(0.52)	(0.67)	(0.43)
Δ*EARN*	-0.271***	-0.264***	-0.258***
	(-11.64)	(-10.94)	(-11.65)
MFNUM	0.002	0.004	0.008
	(0.32)	(0.76)	(1.52)
A_EXP	0.005	0.006	0.008*
	(1.23)	(1.45)	(1.74)
B_ANA	0.004	0.004	0.004
	(1.10)	(1.15)	(0.95)
NoSIC2	0.003	0.002	0.000
	(0.38)	(0.20)	(0.04)
NoFirms	0.000	0.001	0.003
	(-0.00)	(0.12)	(0.41)
Constant	-0.042**	-0.061***	-0.069***
	(-2.05)	(-2.95)	(-3.31)
Adj R^2	0.630	0.624	0.633
Year/Analyst/Ind FE	yes	yes	yes

Observations	22,500	22,500	22,500

Panel A presents the OLS estimation results of the following model:

$$ACCURACY_{ijt} = \beta_0 + \beta_1 SEG_SAMELC_{it} + \beta_2 ESTABNUM_{it} + \beta_3 SEG_SAMELC * ESTABNUM_{it} + \beta_4 SIZE_{it} + \beta_5 EVOL_{it} + \beta_6 SKEW_{it} + \beta_7 LOSS_{it} + \beta_8 BTM_{it} + \beta_9 DISP + \beta_{10} FOLLOW_{it} + \beta_{11} \Delta EARN_{it} + \beta_{12} MFNUM_{it} + \beta_{13} A_EXP_{ijt} + \beta_{14} B_ANA_{ijt} + \beta_{15} NoSIC2_{ijt} + \beta_{16} NoFirms_{ijt} + \varepsilon_{ijt}$$

where *ACCURACY* is (-1) multiplied by the absolute value of the analyst's eps forecast in year t minus the actual eps, divided by the price at the end of the previous fiscal year-end. *SEG_SAMELC* is a dummy variable that is equal to one if all of the firm's 10-K segments are in the same life cycle stage from the 5-stage life cycle using rankings of all segments and zero otherwise. *ESTABNUM* is the natural log of (1 + the number of the firm's segments in year t).

Panel B presents the OLS estimation results of the following model:

$$ACCURACY_{ijt} = \beta_0 + \beta_1 SEG_STG1_PROP_{it} + \beta_2 SEG_STG2_PROP_{it} + \beta_3 SEG_STG3_PROP_{it} + \beta_4 SEG_STG4_PROP_{it} + \beta_5 SIZE_{it} + \beta_6 EVOL_{it} + \beta_7 SKEW_{it} + \beta_8 LOSS_{it} + \beta_9 BTM_{it} + \beta_{10} DISP_{it} + \beta_{11} FOLLOW_{it} + \beta_{12} \Delta EARN_{it} + \beta_{13} MFNUM_{it} + \beta_{14} ESTABNUM_{it} + \beta_{15} A_EXP_{ijt} + \beta_{16} B_ANA_{ijt} + \beta_{17} NoSIC2_{ijt} + \beta_{18} NoFirms_{ijt} + \varepsilon_{ijt}$$

For *SEG_STGn_PROP* where *n* = 1, 2, 3, and 4, we use the following variables. the total segment sales (capex, and employment) of stage *n* segments divided by the total segment sales (capex, and employment) of the firm where stage 1= introduction, stage 2= growth, stage 3= maturity, stage 4= shake-out, and stage 5= decline stage. *TVS* is the total value of shipments, *CAPEX* is the total capex, *EMP* is the total employment, and *PRODHR* is the total production worker hours of an establishment. Other variables are defined in Appendix. *T*-statistics are shown in the parenthesis. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

TABLE 8

Analysts' forecast accuracy and the gap between the 10-K segments and the establishments

	(1)
VARIABLES	
GAP	-0.027***
	(-4.99)
SIZE	-0.003
	(-1.06)
EVOL	0.085
	(1.46)
SKEW	0.005***
	(3.46)
LOSS	-0.078***
	(-7.12)
BTM	0.038***
	(5.69)
DISP	-1.561***
	(-5.31)
FOLLOW	0.011
	(1.57)
Δ*EARN*	-0.257***
	(-11.16)
MFNUM	0.005
	(0.92)
ESTABNUM	0.022***
	(3.10)
A_EXP	0.005
	(1.04)
B_ANA	0.004
	(1.02)
NoSIC2	0.003
	(0.41)
NoFirms	0.000
	(-0.02)
Constant	-0.086***
	(-3.74)
Adj R^2	0.621
Year/Analyst/Ind FE	yes
Observations	22,500

This table presents the OLS estimation results of the following model:

$$ACCURACY_{ijt} = \beta_0 + \beta_1 GAP_{it} + \beta_2 SIZE_{it} + \beta_3 EVOL_{it} + \beta_4 SKEW_{it} + \beta_5 LOSS_{it} + \beta_6 BTM_{it} + \beta_7 DISP_{it} + \beta_8 FOLLOW_{it} + \beta_9 \Delta EARN_{it} + \beta_{10} MFNUM_{it} + \beta_{11} ESTABNUM_{it} + \beta_{12} A_EXP_{ijt} + \beta_{13} B_ANA_{ijt} + \beta_{14} NoSIC2_{ijt} + \beta_{15} NoFirms_{ijt} + \varepsilon_{ijt}$$

where *ACCURACY* is (-1) multiplied by the absolute value of the analyst's eps forecast in year t minus the actual eps, divided by the price at the end of the previous fiscal year-end. *GAP* is the absolute value of the difference between the number of establishments and the number of segments of a firm. Other variables are defined in Appendix. *T*-statistics are shown in the parenthesis. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.

TABLE 9

Management forecast accuracy and the proportion of establishments in the early life-cycle stages

	(1)	(2)	(3)	(4)
VARIABLES	*TVS*	*CAPEX*	*EMP*	*PRODHR*
STG1_PROP	-0.034**	-0.032*	-0.029*	-0.026*
	(-2.17)	(-1.87)	(-1.74)	(-1.88)
STG2_PROP	-0.044**	-0.026	-0.03	-0.028*
	(-2.42)	(-1.39)	(-1.62)	(-1.82)
STG3_PROP	-0.029	-0.021	-0.024	-0.026
	(-1.50)	(-1.09)	(-1.15)	(-1.46)
STG4_PROP	-0.024	-0.016	-0.021	-0.022
	(-1.27)	(-0.86)	(-1.07)	(-1.28)
LOSS	-0.045***	-0.043**	-0.043***	-0.044***
	(-2.72)	(-2.62)	(-2.71)	(-2.76)
SIZE	0.001	0.002	0.001	0.001
	(0.41)	(0.79)	(0.42)	(0.24)
EVOL	0.116***	0.113***	0.105***	0.095***
	(3.81)	(3.60)	(3.63)	(3.28)
SKEW	0.003*	0.003	0.003	0.002
	(1.73)	(1.63)	(1.41)	(1.34)
BTM	-0.029***	-0.017*	-0.018*	-0.018**
	(-3.58)	(-1.91)	(-1.92)	(-2.04)
DISP	-1.226***	-0.631***	-0.587***	-0.602***
	(-5.10)	(-3.01)	(-2.73)	(-2.81)
FOLLOW	-0.005	-0.002	-0.003	-0.003
	(-1.37)	(-0.55)	(-0.79)	(-0.79)
Δ*EARN*	-0.288***	-0.288***	-0.280***	-0.275***
	(-4.99)	(-4.72)	(-4.65)	(-4.74)
MFNUM	-0.005	-0.003	-0.003	-0.003
	(-1.08)	(-0.73)	(-0.78)	(-0.76)
ESTABNUM	0.001	0.001	0.001	0.001
	(1.25)	(0.85)	(1.11)	(1.21)
Constant	0.038	0.01	0.019	0.022
	(1.50)	(0.46)	(0.74)	(0.91)
Adj R^2	0.681	0.646	0.641	0.640
Year/Ind FE	yes	yes	yes	yes
Observations	650	650	650	650

This table presents the OLS estimation results of the following model:

$$MFACCURACY_{ijt} = \beta_0 + \beta_1 STG1_PROP_{it} + \beta_2 STG2_PROP_{it} + \beta_3 STG3_PROP_{it} + \beta_4 STG4_PROP_{it} + \beta_5 SIZE_{it} + \beta_6 EVOL_{it} + \beta_7 SKEW_{it} + \beta_8 LOSS_{it} + \beta_9 BTM_{it} + \beta_{10} DISP_{it} + \beta_{11} FOLLOW_{it} + \beta_{12} \Delta EARN_{it} + \beta_{13} MFNUM_{it} + \beta_{14} ESTABNUM_{it} + \beta_{15} A_EXP_{ijt} + \beta_{16} B_ANA_{ijt} + \beta_{17} NoSIC2_{ijt} + \beta_{18} NoFirms_{ijt} + \varepsilon_{ijt}$$

where *MFACCURACY* is (-1) multiplied by the absolute value of the management's eps forecast in year t minus the actual eps, divided by the price at the end of the previous fiscal year-end. For *STGn_PROP* where *n* = 1, 2, 3, and 4, *STGn_TVS*, *STGn_CAPEX*, *STGn_EMP*, and *STGn_PRODHR* are used in columns (1), (2), (3), and (4), respectively. *STGn_TVS* (*STGn_CAPEX*; *STGn_EMP*; *STGn_PRODHR*) is the total *TVS* (*CAPEX*; *EMP*; *PRODHR*) of stage *n* establishments divided by the total *TVS* (*CAPEX*; *EMP*; *PRODHR*) of establishments of a firm, where the life-cycle stage is estimated based on the life-cycle descriptors' rankings for all establishments. *TVS* is the total value of shipments, *CAPEX* is the total capex, *EMP* is the total employment, and *PRODHR* is the total production worker hours of an establishment. Other variables are defined in Appendix. *T*-statistics are shown in the parenthesis. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels (two-tailed), respectively.